

3.7 How Science Works

How Science Works is an underpinning set of concepts and is the means whereby students come to understand how scientists investigate scientific phenomena in their attempts to explain the world about us. Moreover, *How Science Works* recognises the contribution scientists have made to their own disciplines and to the wider world.

Further, it recognises that scientists may be influenced by their own beliefs and that these can affect the way in which they approach their work. Also, it acknowledges that scientists can and must contribute to debates about the uses to which their work is put and how their work influences decision-making in society.

In general terms, it can be used to promote students' skills in solving scientific problems by developing an understanding of:

- the concepts, principles and theories that form the subject content
- the procedures associated with the valid testing of ideas and, in particular, the collection, interpretation and validation of evidence
- the role of the scientific community in validating evidence and also in resolving conflicting evidence.

As students become proficient in these aspects of *How Science Works*, they can also engage with the place and contribution of science in the wider world. In particular, students will begin to recognise:

- the contribution that scientists can make to decision-making and the formulation of policy
- the need for regulation of scientific enquiry and how this can be achieved
- how scientists can contribute legitimately to debates about those claims which are made in the name of science.

An understanding of *How Science Works* is a requirement for this specification and is set out in the following points which are taken direct from the *GCE AS and A Level subject criteria for science subjects*. Each point is expanded in the context of Chemistry. The specification references given illustrate where the example is relevant and could be incorporated.

A	Use theories, models and ideas to develop and modify scientific explanations
	Scientists use theories and models to attempt to explain observations. These theories or models can form the basis for scientific experimental work. Scientific progress is made when validated evidence is found that supports a new theory or model.
	<i>Examples in this specification include:</i> - the use of ionisation energy plots as evidence for electron arrangement in shells and subshells (AS Unit 1, 3.1.1) - experiments with cells confirm that electrons are transferred in redox reactions (A2 Unit 5, 3.5.3)
B	Use knowledge and understanding to pose scientific questions, define scientific problems, present scientific arguments and scientific ideas
	Scientists use their knowledge and understanding when observing objects and events, in defining a scientific problem and when questioning the explanations of themselves or of other scientists. Scientific progress is made when scientists contribute to the development of new ideas, materials and theories.
	<i>Examples in this specification include:</i> - explanation of the origin of the hole in the ozone layer (AS Unit 2, 3.2.8) - Entropy as a concept to explain spontaneous reactions (A2 Unit 5, 3.5.2)
C	Use appropriate methodology, including ICT, to answer scientific questions and solve scientific problems
	Observations ultimately lead to explanations in the form of hypotheses. In turn, these hypotheses lead to predictions that can be tested experimentally. Observations are one of the key links between the 'real world' and the abstract ideas of science. Once an experimental method has been validated, it becomes a protocol that is used by other scientists. ICT can be used to speed up, collect, record and analyse experimental data.
	<i>Examples in this specification include:</i> Many opportunities permeating throughout the Practical and Investigative Skills units (Unit 3 and Unit 6)
D	Carry out experimental and investigative activities, including appropriate risk management, in a range of contexts
	Scientists perform a range of experimental skills that include manual and data skills (tabulation, graphical skills etc). Scientists should select and use equipment that is appropriate when making accurate measurements and should record these measurements methodically. Scientists carry out experimental work in such a way as to minimise the risk to themselves, to others and to the materials, including organisms, used.
	<i>Examples in this specification include:</i> Many opportunities permeating throughout the Practical and Investigative Skills units (Unit 3 and Unit 6)

E	Analyse and interpret data to provide evidence, recognising correlations and causal relationships Scientists look for patterns and trends in data as a first step in providing explanations of phenomena. The degree of uncertainty in any data will affect whether alternative explanations can be given for the data. Anomalous data are those measurements that fall outside the normal, or expected, range of measured values. Decisions on how to treat anomalous data should be made only after examination of the event. In searching for causal links between factors, scientists propose predictive theoretical models that can be tested experimentally. When experimental data confirm predictions from these theoretical models, scientists become confident that a causal relationship exists. <i>Examples in this specification include:</i> - the use of enthalphy of combustion data for a range of alcohols in the development of the idea of mean bond enthalpies (AS Unit 2, 3.2.1) - the recognition that entropy change is an important factor in determining the direction of spontaneous reaction (A2 Unit 5, 3.5.1 and 3.5.5)
F	Evaluate methodology, evidence and data, and resolve conflicting evidence The validity of new evidence, and the robustness of conclusions that stem from them, is constantly questioned by scientists. Experimental methods must be designed adequately to test predictions. Solutions to scientific problems are often developed when different research teams produce conflicting evidence. Such evidence is a stimulus for further scientific investigation, which involves refinements of experimental technique or development of new hypotheses. <i>Examples in this specification include:</i> - the importance of bond polarity and carbon-halogen bond enthalpy as factors in determining the rate of hydrolysis of haloalkanes (AS Unit 1, 3.1.1) - the use of thermochemical evidence from enthalpies of hydrogenation enthalpies as support for the structure of benzene (A2 Unit 4, 3.4.6)
G	Appreciate the tentative nature of scientific knowledge Scientific explanations are those that are based on experimental evidence which is supported by the scientific community. Scientific knowledge changes when new evidence provides a better explanation of scientific observations. <i>Examples in this specification include:</i> - the increase in greenhouse gases that may cause global warming (AS Unit 1, 3.1.6) - the reactions of metal-aqua ions acting as Lewis bases (A2 Unit 5, 3.5.5)
H	Communicate information and ideas in appropriate ways using appropriate terminology By sharing the findings of their research, scientists provide the scientific community with opportunities to replicate and further test their work, thus either confirming new explanations or refuting them. Scientific terminology avoids confusion amongst the scientific community, enabling better understanding and testing of scientific explanations. <i>Examples in this specification include:</i> - IUPAC rules for naming organic compounds have been adopted globally (AS Unit 1, 3.1.5) - experiments with cells confirm that electrons are transferred in redox reactions (A2, Unit 5, 3.5.3)

I	Consider applications and implications of science and appreciate their associated benefits and risks
	Scientific advances have greatly improved the quality of life for the majority of people. Developments in technology, medicine and materials continue to further these improvements at an increasing rate.
	Scientists can predict and report on some of the beneficial applications of their experimental findings. Scientists evaluate, and report on, the risks associated with the techniques they develop and applications of their findings.
	<i>Examples in this specification include:</i> • <i>the benefits and risks of using chlorine in water treatment (AS Unit 2, 3.2.5)</i> • <i>The use of hydrogen-oxygen fuel cells as a source of energy and the hazards associated with their use (A2 Unit 5, 3.5.4)</i>
J	Consider ethical issues in the treatment of humans, other organisms and the environment
	Scientific research is funded by society, either through public funding or through private companies that obtain their income from commercial activities. Scientists have a duty to consider ethical issues associated with their findings.
	Individual scientists have ethical codes that are often based on humanistic, moral and religious beliefs. Scientists are self-regulating and contribute to decision making about what investigations and methodologies should be permitted.
	<i>Examples in this specification include:</i> • <i>the environmental and economic advantages of recycling scrap metal (AS Unit 2, 3.2.7)</i> • <i>how the relative biodegradability of polymers affects their disposal or reuse (A2, Unit 4, 3.4.9)</i>
K	Appreciate the role of the scientific community in validating new knowledge and ensuring integrity
	The findings of scientists are subject to peer review before being accepted for publication in a reputable scientific journal.
	The interests of the organisations that fund scientific research can influence the direction of research. In some cases the validity of those claims may also be influenced
	<i>Examples in this specification include:</i> • <i>the identification of acid rain as a problem (AS Unit 2, 3.2.7)</i> • <i>the production of biofuels as carbon neutral fuels (A2 Unit 4, 3.4.4)</i>
L	Appreciate the ways in which society uses science to inform decision making
	Scientific findings and technologies enable advances to be made that have potential benefit for humans.
	In practice, the scientific evidence available to decision makers may be incomplete. Decision makers are influenced in many ways, including by their prior beliefs, their vested interests, special interest groups, public opinion and the media, as well as by expert scientific evidence
	<i>Examples in this specification include:</i> • <i>the identification of the release of CO and NO as a problem and the development of catalytic converters to counteract this (AS Unit 2, 3.2.8)</i> • <i>the different properties of enantiomers can give rise to different chemical reactions in the body, for example the optical isomers of thalidomide (A2 Unit 4, 3.4.4)</i>